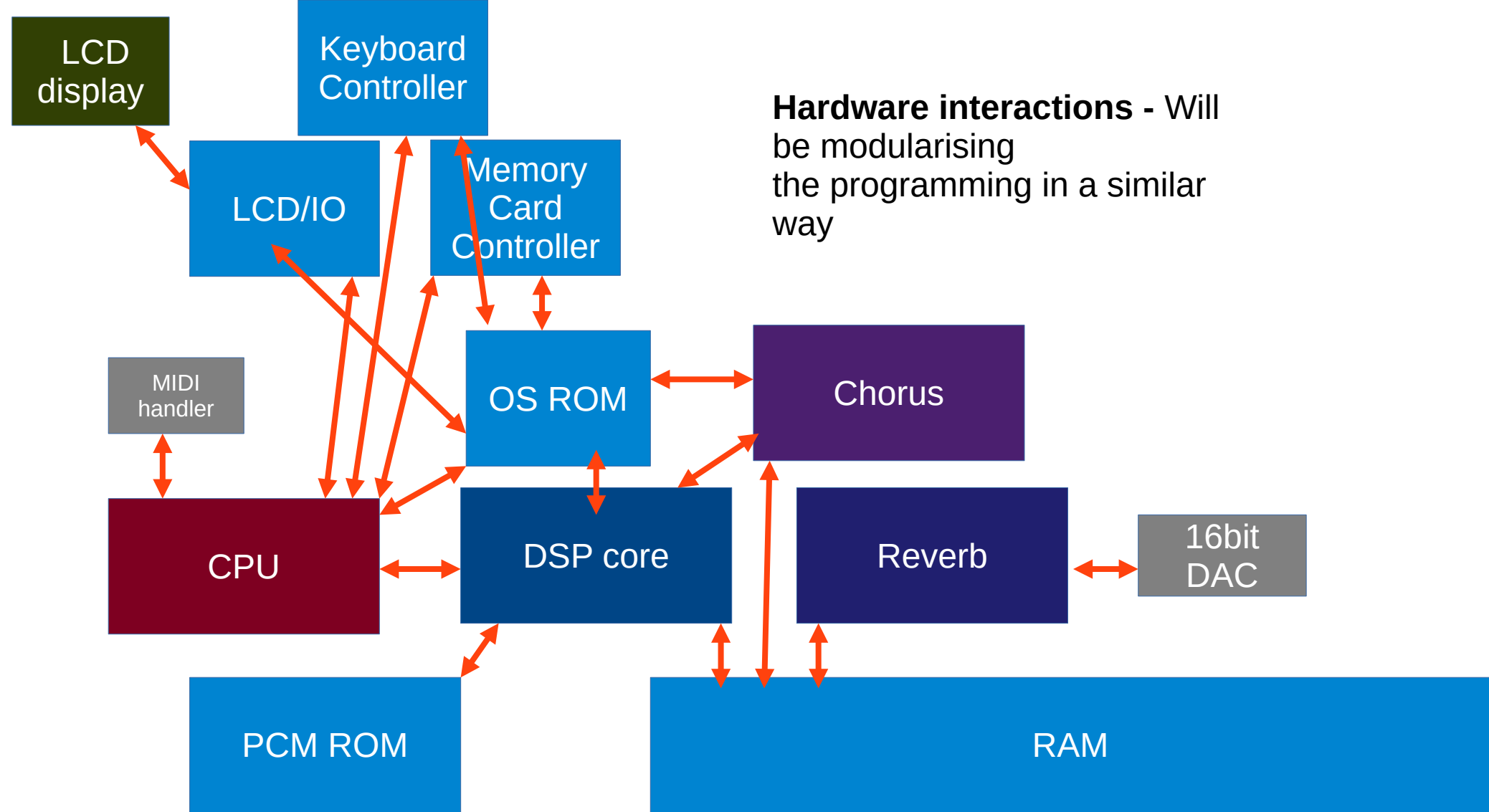


xD-69 Hardware Description



This amazing logo was designed by someone called kilésengati on Discord :)



CPU: NEC uPD78312 – 8/16 bit embedded microcontroller

U_1

NEC 78k/3 series microcontroller for embedded devices. CISC(Complex instruction set computer) processor, little endian.
No current ROM dump. Has a PROM that contains opcode stuff I think
Serial MIDI transmission uses RXD and TXD CPU pins.

1	P0.0	VDD	64
2	P0.1	AD7	63
3	P0.2	AD6	62
4	P0.3	AD5	61
5	P0.4	AD4	60
6	P0.5	AD3	59
7	P0.6	AD2	58
8	P0.7	AD1	57
9	P1.0	AD0	56
10	P1.1	ALE	55
11	P1.2	WR	54
12	P1.3	RD	53
13	P1.4	RESET	52
14	P1.5	EA	51
15	P1.6	A15	50
16	P1.7	A14	49
17	NMI	A13	48
18	INT0	A12	47
19	INT1	A11	46
20	INT2	A10	45
21	TXD	A9	44
22	RXD	A8	43
23	SCK	P3.7	42
24	CTS	P3.6	41
25	RFSH	P3.5	40
26	P3.0	P3.4	39
27	P3.1	AVss	38
28	P3.2	AVref	37
29	P3.3	AN3	36
30	X1	AN2	35
31	X2	AN1	34
32	VSS	AN0	33

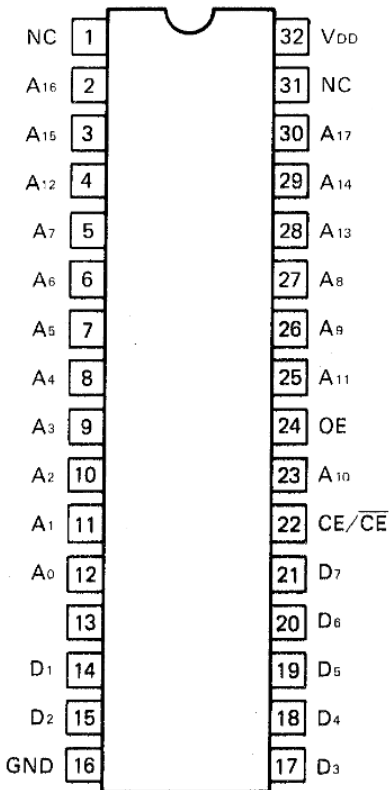
Pin No.	Pin Name	Input/Output	Description	Pin No.	Pin Name	Input/Output	Description
1	P0.0	O	Port 0 – Bit 0	33	AN0	I	Analog input
2	P0.1	O	Port 0 – Bit 1	34	AN1	I	Analog input
3	P0.2	O	Port 0 – Bit 2	35	AN2	I	Analog input
4	P0.3	O	Port 0 – Bit 3	36	AN3	I	Analog input
5	P0.4	O	Port 0 – Bit 4	37	AVREF	O	Reference voltage input for A/D converter
6	P0.5	O	Port 0 – Bit 5	38	AVSS	-	Analog VSS/ground
7	P0.6	O	Port 0 – Bit 6	39	P3.4	I/O(NC)	Port 3 – Bit 4
8	P0.7	O	Port 0 – Bit 7	40	P3.5	I/O(NC)	Port 3 – Bit 5
9	P1.0	I/O	Port 1 – Bit 0	41	P3.6	I/O	Port 3 – Bit 6
10	P1.1	I/O	Port 1 – Bit 1	42	P3.7	I/O	Port 3 – Bit 7
11	P1.2	I/O	Port 1 – Bit 2	43	A8	O	Address bus
12	P1.3	I/O	Port 1 – Bit 3	44	A9	O	Address bus
13	P1.4	I/O	Port 1 – Bit 4	45	A10	O	Address bus
14	P1.5	I/O	Port 1 – Bit 5	46	A11	O	Address bus
15	P1.6	I/O	Port 1 – Bit 6	47	A12	O	Address bus
16	P1.7	I/O(NC)	Port 1 – Bit 7	48	A13	O	Address bus
17	NMI	I	Non-maskable interrupt	49	A14	O	Address bus
18	INT0	I	Interrupt from externals	50	A15	O	Address bus
19	INT1	I	Interrupt from externals	51	EA	I	Enables use of external memory
20	INT2	I	Interrupt from externals	52	RESET	I	External reset input
21	TXD	O	Transmitted serial data output	53	RD	O	Read strobe output
22	RXD	I	Received serial data input	54	WR	O	Write strobe
23	SCK	O(NC)	Serial transmit clock output	55	ALE	O	Address latch enable
24	CTS	I/O	Clear to send input	56	AD0	I/O	Address/data bus
25	RFSH	O(NC)	Refresh pulse output	57	AD1	I/O	Address/data bus
26	P3.0	I	Port 3 – Bit 0	58	AD2	I/O	Address/data bus
27	P3.1	I	Port 3 – Bit 1	59	AD3	I/O	Address/data bus
28	P3.2	I	Port 3 – Bit 2	60	AD4	I/O	Address/data bus
29	P3.3	I	Port 3 – Bit 3	61	AD5	I/O	Address/data bus
30	X1	I	Crystal	62	AD6	I/O	Address/data bus
31	X2	I	Crystal	63	AD7	I/O	Address/data bus
32	VSS	-	Ground	64	VDD	-	Power

PCM storage ROM – contains PCM partials for the preset playback. Has been ROM dumped. Roland owns the samples so you need to provide your own copy

Read-Only Memories

OS ROM - This is where the D-50 runs its code from for the D-50 operating system. I have provided the disassembly in another folder. I managed to disassemble the binary using MAME disasm command line

PCM ROM A/B TC532000



TOP VIEW

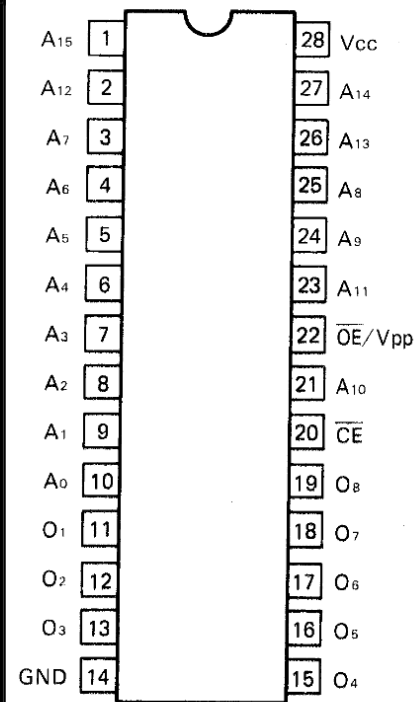
TC532000 ROM (PCM)

Pin No.	Pin Name	Input/Output	Description
1	NC	-	No connection
2	A16	I	Address inputs
3	A15	I	Address inputs
4	A12	I	Address inputs
5	A7	I	Address inputs
6	A6	I	Address inputs
7	A5	I	Address inputs
8	A4	I	Address inputs
9	A3	I	Address inputs
10	A2	I	Address inputs
11	A1	I	Address inputs
12	A0	I	Address inputs
13	D0	O	Data outputs
14	D1	O	Data outputs
15	D2	O	Data outputs
16	GND	-	Ground
17	D3	O	Data outputs
18	D4	O	Data outputs
19	D5	O	Data outputs
20	D6	O	Data outputs
21	D7	O	Data outputs
22	CE/CE	I	Chip enable input
23	A10	I	Address inputs
24	OE	I	Output enable input
25	A11	I	Address inputs
26	A9	I	Address inputs
27	A8	I	Address inputs
28	A13	I	Address inputs
29	A14	I	Address inputs
30	A17	I	Address inputs
31	NC	-	No connection
32	VDD	-	Power

MBM27C512 ROM (OS)

Pin No.	Pin Name	Input/Output	Description
1	A15	I	Address input
2	A12	I	Address input
3	A7	I	Address input
4	A6	I	Address input
5	A5	I	Address input
6	A4	I	Address input
7	A3	I	Address input
8	A2	I	Address input
9	A1	I	Address input
10	A0	I	Address input
11	O1	O	Data output
12	O2	O	Data output
13	O3	O	Data output
14	GND	-	Ground
15	O4	O	Data output
16	O5	O	Data output
17	O6	O	Data output
18	O7	O	Data output
19	O8	O	Data output
20	CE	I	Chip enable input
21	A10	I	Address input
22	OE/VPP	I	Output enable input
23	A11	I	Address input
24	A9	I	Address input
25	A8	I	Address input
26	A13	I	Address input
27	A14	I	Address input
28	VCC	-	Power

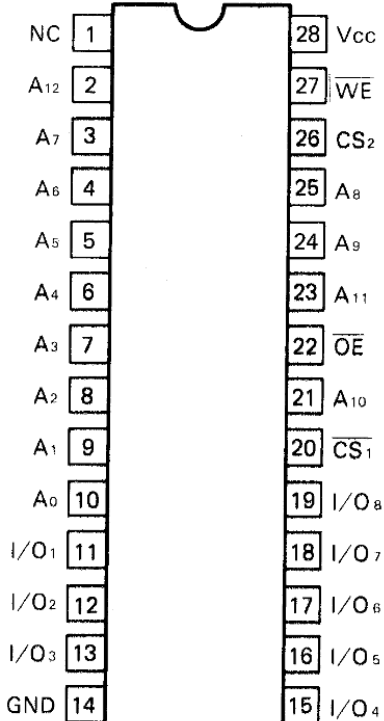
EP ROM MBM27C512



TOP VIEW

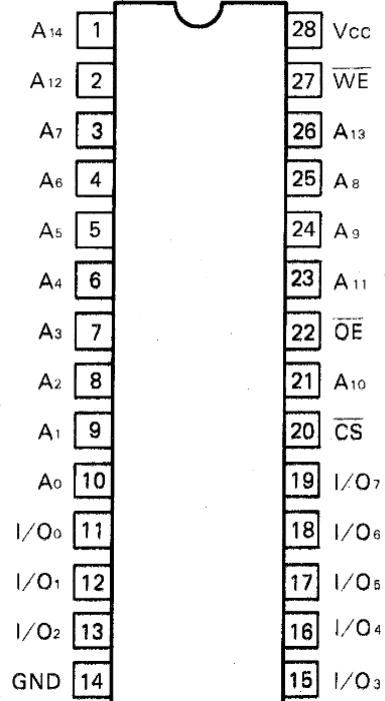
Random-Access Memories

**S RAM
HM6264ASP**



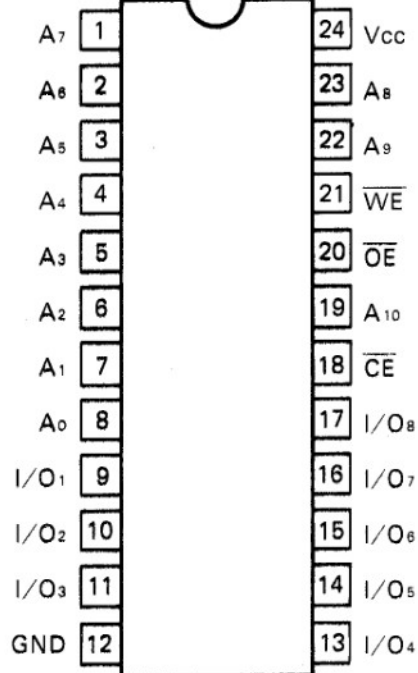
TOP VIEW

**S RAM
HM62256LP**



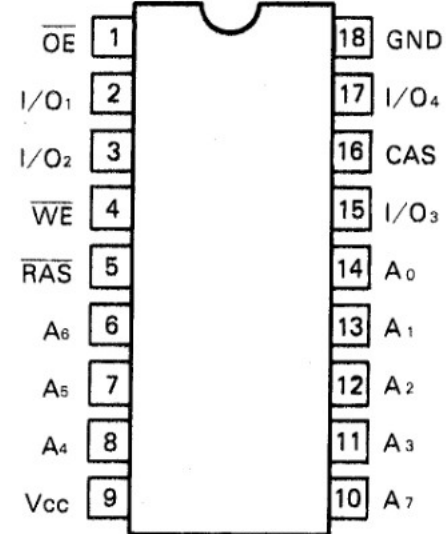
TOP VIEW

**S RAM
LC3517AS**



TOP VIEW

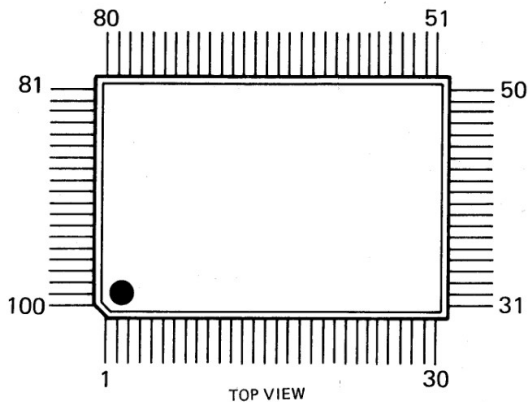
**D RAM
 μ PD41416**



TOP VIEW

Target Custom ASIC – Chorus DSP

**CHORUS CUSTOM IC
MB87137**

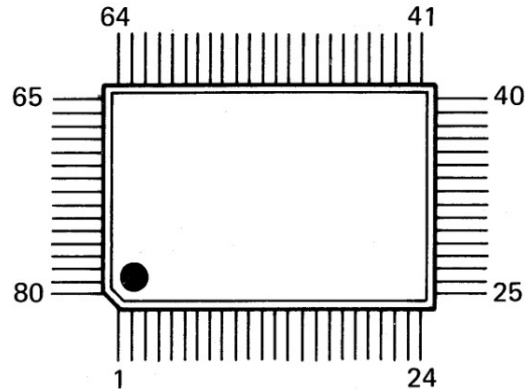


Please see:
<https://github.com/pxlsr/FujiCells>
for a detailed and comprehensive
documentation of the ASICs

Pin No.	Pin Name	Input/Output	Description	Pin No.	Pin Name	Input/Output	Description
1	RES	I	Reset input	51	O8	O	Data input
2	E	I	Chip enable input	52	O9	O	Data input
3	VDD	-	+5v	53	VDD	-	+5v
4	CS	I	Chip Select input	54	O10	O	Data input
5	RW	I	Write Pulse input	55	O11	O	Data input
6	RD	I	Read Pulse input	56	O12	O	Data input
7	CS	I	Chip select input	57	O13	O	Data input
8	A0	I	Connect to CPU address bus	58	O14	O	Data input
9	A1	I	Connect to CPU address bus	59	O15	O	Data input
10	A2	I	Connect to CPU address bus	60	RA0	O	Connects to SRAM address bus RA13 not used
11	D0	I/O	Connect to CPU data bus	61	WE	O	SRAM write pulse output
12	D1	I/O	Connect to CPU data bus	62	RA1	O	Connects to SRAM address bus RA13 not used
13	D2	I/O	Connect to CPU data bus	63	RA2	O	Connects to SRAM address bus RA13 not used
14	D3	I/O	Connect to CPU data bus	64	RA3	O	Connects to SRAM address bus RA13 not used
15	VSS	-	GND	65	VSS	-	GND
16	D4	I/O	Connect to CPU data bus	66	RA4	O	Connects to SRAM address bus RA13 not used
17	D5	I/O	Connect to CPU data bus	67	RA5	O	Connects to SRAM address bus RA13 not used
18	D6	I/O	Connect to CPU data bus	68	RA6	O	Connects to SRAM address bus RA13 not used
19	D7	I/O	Connect to CPU data bus	69	RA7	O	Connects to SRAM address bus RA13 not used
20	DOE	I	Data out enable input	70	RA8	O	Connects to SRAM address bus RA13 not used
21	INCK	I	Input data latch clock input	71	OE	O	SRAM out enable output
22	SIN	I	Sync input	72	RA9	O	Connects to SRAM address bus RA13 not used
23	SOUT	O	Sync output	73	RA10	O	Connects to SRAM address bus RA13 not used
24	LRS	I	L/R select input	74	RA11	O	Connects to SRAM address bus RA13 not used
25	I0	I	Data input	75	CE	O	SRAM chip enable output
26	I1	I	Data input	76	RA12	O	Connects to SRAM address bus RA13 not used
27	I2	I	Data input	77	PD7	I/O	Connect to SRAM data bus
28	VDD	-	+5v	78	VDD	-	+5v
29	I3	I	Data input	79	RA13	O	Connects to SRAM address bus RA13 not used
30	I4	I	Data input	80	PD6	I/O	Connect to SRAM data bus
31	I5	I	Data input	81	PD5	I/O	Connect to SRAM data bus
32	I6	I	Data input	82	PD4	I/O	Connect to SRAM data bus
33	I7	I	Data input	83	PD3	I/O	Connect to SRAM data bus
34	I8	I	Data input	84	PD2	I/O	Connect to SRAM data bus
35	I9	I	Data input	85	PD1	I/O	Connect to SRAM data bus
36	I10	I	Data input	86	PD0	I/O	Connect to SRAM data bus
37	I11	I	Data input	87	VSS	-	GND
38	I12	I	Data input	88	X1	I	Master clock input
39	I13	I	Data input	89	X2	O	not used
40	VSS	-	GND	90	VSS	-	GND
41	I14	I	Data input	91	ROMT	I	No description
42	I15	I	Data input	92	RAMT	I	No description
43	O0	O	Data input	93	CTRT	I	No description
44	O1	O	Data input	94	THRU	I	No description
45	O2	O	Data input	95	ECTL	I	External control select input
46	O3	O	Data input	96	ADDA	I	No description
47	O4	O	Data input	97	OFST	I	offset binary select input
48	O5	O	Data input	98	PSFT	I	No description
49	O6	O	Data input	99	LHLD	O	Signal output for S/H – not used
50	O7	O	Data input	100	RHLD	O	Signal output for S/H – not used

Target Custom ASIC – Reverb DSP

REVERB CUSTOM IC MB87126-006



TOP VIEW

Pin No.	Pin Name	Input/Output	Description	Pin No.	Pin Name	Input/Output	Description
1	DC0	O	Data output for chorus chip and DAC	51	DR11	I/O	Connect to RAM data bus, Synth and Chorus data input
2	DC1	O	Data output for chorus chip and DAC	52	VSS	-	GND
3	STR1	I	Pulled low	53	DR12	I/O	Connect to RAM data bus, Synth and Chorus data input
4	DIN	I	Pulled low	54	DR13	I/O	Connect to RAM data bus, Synth and Chorus data input
5	CLEA	I	Pulled low	55	DR14	I/O	Connect to RAM data bus, Synth and Chorus data input
6	RD0	O	Control output for enable and for S/H and Lower four bit D/A conversion	56	DR15	I/O	Connect to RAM data bus, Synth and Chorus data input
7	RD1	O	Control output for enable and for S/H and Lower four bit D/A conversion	57	DR16	I/O	Connect to RAM data bus, Synth and Chorus data input
8	RD2	O	Control output for enable and for S/H and Lower four bit D/A conversion	58	DR17	I/O	Connect to RAM data bus, Synth and Chorus data input
9	RD3	O	Control output for enable and for S/H and Lower four bit D/A conversion	59	DR18	I/O	Connect to RAM data bus, Synth and Chorus data input
10	RD4	O	Control output for enable and for S/H and Lower four bit D/A conversion	60	DR19	I/O	Connect to RAM data bus, Synth and Chorus data input
11	RSET	I	Pulled low	61	DR20	I/O	Connect to RAM data bus, Synth and Chorus data input
12	VSS	-	GND	62	DR21	I/O	Connect to RAM data bus, Synth and Chorus data input
13	SLRQ	I	Pulled low	63	DR22	I/O	Connect to RAM data bus, Synth and Chorus data input
14	MSCK	I	Master clock input	64	DR23	I/O	Connect to RAM data bus, Synth and Chorus data input
15	VSS	-	GND	65	VSS	-	GND
16	SLCK	O	Not used	66	DC2	O	Data output for chorus chip and DAC
17	TEST	I	Pulled low	67	DC3	O	Data output for chorus chip and DAC
18	TMB	O	Time base signal output	68	DC4	O	Data output for chorus chip and DAC
19	VDD	-	+5V	69	DC5	O	Data output for chorus chip and DAC
20	LOAD	O	Sync signal output	70	DC6	O	Data output for chorus chip and DAC
21	SYNC	I	Sync signal input	71	DC7	O	Data output for chorus chip and DAC
22	INCK	I	Data latch clock input for initialization	72	DC8	O	Data output for chorus chip and DAC
23	ERCL	I	Busy reset output	73	VDD	-	+5V
24	BUSY	O	Serial data transfer error output (parity check)	74	DC9	O	Data output for chorus chip and DAC
25	SXD	I	Serial data input	75	DC10	O	Data output for chorus chip and DAC
26	SCK	I	Serial data read-in clock input	76	DC11	O	Data output for chorus chip and DAC
27	DA0	O	Connect to RAM address bus	77	DC12	O	Data output for chorus chip and DAC
28	DA1	O	Connect to RAM address bus	78	DC13	O	Data output for chorus chip and DAC
29	DA2	O	Connect to RAM address bus	79	DC14	O	Data output for chorus chip and DAC
30	DA3	O	Connect to RAM address bus	80	DC15	O	Data output for chorus chip and DAC
31	DA4	O	Connect to RAM address bus				
32	DA5	O	Connect to RAM address bus				
33	VDD	-	+5V				
34	DA6	O	Connect to RAM address bus				
35	DA7	O	Connect to RAM address bus				
36	VSS	-	GND				
37	RAS	O	Row address strobe output				
38	WE	O	DRAM write pulse output				
39	CAS	O	Column address strobe output				
40	DR0	I/O	Connect to RAM data bus, Synth and Chorus data input				
41	DR1	I/O	Connect to RAM data bus, Synth and Chorus data input				
42	DR2	I/O	Connect to RAM data bus, Synth and Chorus data input				
43	DR3	I/O	Connect to RAM data bus, Synth and Chorus data input				
44	DR4	I/O	Connect to RAM data bus, Synth and Chorus data input				
45	DR5	I/O	Connect to RAM data bus, Synth and Chorus data input				
46	DR6	I/O	Connect to RAM data bus, Synth and Chorus data input				
47	DR7	I/O	Connect to RAM data bus, Synth and Chorus data input				
48	DR8	I/O	Connect to RAM data bus, Synth and Chorus data input				
49	DR9	I/O	Connect to RAM data bus, Synth and Chorus data input				
50	DR10	I/O	Connect to RAM data bus, Synth and Chorus data input				

Please see:
<https://github.com/pxlsr/FujiCells>
 for a detailed and comprehensive
 documentation of the ASICs

Target Custom ASIC – Synth (DSP core)

**SYNTH CUSTOM IC
MB87136**

34	33	32	31	30	29	28	27	26	25	24	23
35	72	71	70	69	68	67	66	65	64	63	22
36	73				86	85				62	21
37	74									61	20
38	75									60	19
39	76	87							84	59	18
40	77	88							83	58	17
41	78									57	16
42	79	INDEX PIN								56	15
43	80				81	82				55	14
44	45	46	47	48	49	50	51	52	53	54	13
1	2	3	4	5	6	7	8	9	10	11	12

TOP VIEW

Pin No.	Pin Name	Input/Output	Description
1	CS	I	Chip select
2	A0	I	Connect to CPU address bus
3	A1	I	Connect to CPU address bus
4	A2	I	Connect to CPU address bus
5	A3	I	Connect to CPU address bus
6	A4	I	Connect to CPU address bus
7	D0	I/O	Connect to CPU data bus
8	D1	I/O	Connect to CPU data bus
9	D2	I/O	Connect to CPU data bus
10	D3	I/O	Connect to CPU data bus
11	PD0	I	Connect to ROM data bus
12	PD1	I	Connect to ROM data bus
13	PD2	I	Connect to ROM data bus
14	PD3	I	Connect to ROM data bus
15	RA0	O	Connect to ROM address bus
16	RA1	O	Connect to ROM address bus
17	RA2	O	Connect to ROM address bus
18	RA3	O	Connect to ROM address bus
19	RA4	O	Connect to ROM address bus
20	RA5	O	Connect to ROM address bus
21	RA6	O	Connect to ROM address bus
22	RA7	O	Connect to ROM address bus
23	RA8	O	Connect to ROM address bus
24	RA9	O	Connect to ROM address bus
25	RA10	O	Connect to ROM address bus
26	RA11	O	Connect to ROM address bus
27	O0	O	Data output
28	O1	O	Data output
29	O2	O	Data output
30	O3	O	Data output
31	O4	O	Data output
32	O5	O	Data output
33	O6	O	Data output
34	O7	O	Data output
35	O8	O	Data output
36	SH0	O	Not used
37	SH1	O	Not used
38	-	-	Not used
39	X1	I/O	Crystal input (32.768mhz)
40	32M	O	The same frequency as master clock
41	-	-	-
42	SYI	I	Sync signal input
43	WR	I	Write pulse input
44	INT	O	Interrupt output
45	OE	I	Output enable input

Pin No.	Pin Name	Input/Output	Description
46	A5	I	Connect to CPU address bus
47	A6	I	Connect to CPU address bus
48	A7	I	Connect to CPU address bus
49	A8	I	Connect to CPU address bus
50	D4	I/O	Connect to CPU data bus
51	D5	I/O	Connect to CPU data bus
52	D6	I/O	Connect to CPU data bus
53	D7	I/O	Connect to CPU data bus
54	PD4	I	Connect to ROM data bus
55	PD5	I	Connect to ROM data bus
56	PD6	I	Connect to ROM data bus
57	PD7	I	Connect to ROM data bus
58	RA12	O	Connect to ROM address bus
59	RA13	O	Connect to ROM address bus
60	RA14	O	Connect to ROM address bus
61	RA15	O	Connect to ROM address bus
62	RA16	O	Connect to ROM address bus
63	RA17	O	Connect to ROM address bus
64	RA18	O	Connect to ROM address bus
65	RA19	O	Connect to ROM address bus
66	O9	O	Data output
67	O10	O	Data output
68	O11	O	Data output
69	O12	O	Data output
70	O13	O	Data output
71	O14	O	Data output
72	O15	O	Data output
73	SH2	O	Not used
74	SH3	O	Not used
75	-	-	Not used
76	X2	I/O	Crystal input
77	16M	O	Output frequency is half of master clock
78	CKIN	I	Input frequency is a combination of the master clock and one half of master clock
79	-	-	Not used
80	RD	I	Read pulse input
81	VSS	-	GND
82	VDD	-	+5V
83	VDD	-	+5V
84	VSS	-	GND
85	VSS	-	GND
86	VDD	-	+5V
87	VDD	-	+5V
88	VSS	-	GND

Please see:

<https://github.com/pxlsr/FujiCells>

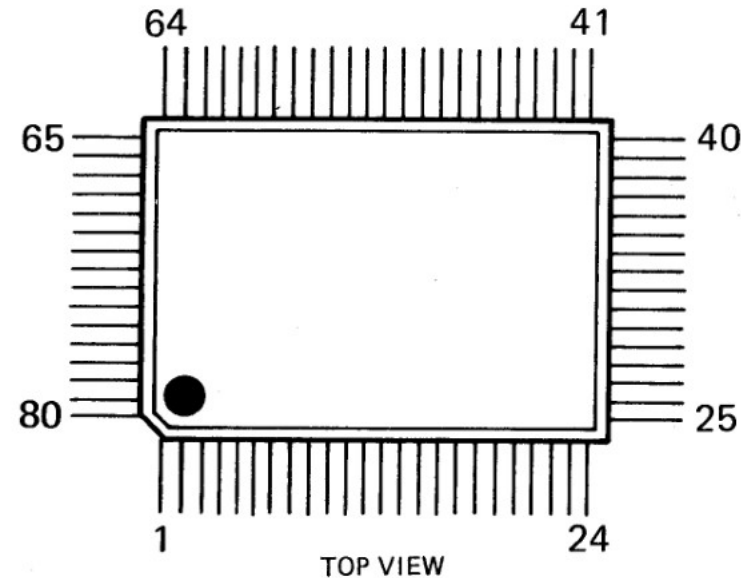
for a detailed and comprehensive documentation of the ASICs

Target Custom Gate Array – LCD controller with possible peripheral control logic?

LCD display unit is a 40x2 Sharp LM402B02(Most likely interchangeable) (Pinout looks compatible with modern alternatives)

Likely being driven by a Hitachi HD44780 LCD controller

GATE ARRAY HG61H25B18F

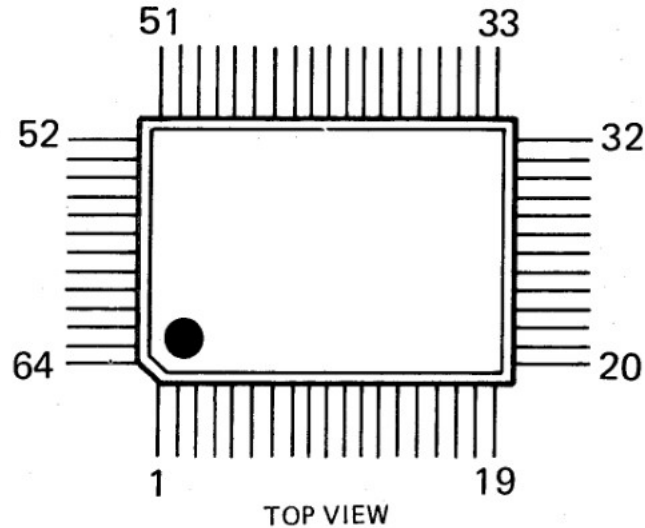


PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O
1	SYNT2	O (NC)	21	ALE	I	41	EC	O	61	R2	I
2	IRAM	O (NC)	22	WR	I	42	O0	O	62	R3	I
3	RAM	O	23	RD	I	43	O1	O	63	R4	I
4	A7	O	24	RESET	I	44	O2	O	64	R5	I
5	A6	O	25	A15	I	45	O3	O	65	R6	I
6	A5	O	26	A14	I	46	O4	O	66	R7	I
7	A4	O	27	A13	I	47	O5	O	67	CORUS	O
8	A3	O	28	A12	I	48	O6	O	68	SCK	O
9	A2	O	29	A11	I	49	O7	O	69	SXD	O
10	A1	O	30	A10	I	50	S0	O	70	BUSY	I
11	A0	O	31	A9	I	51	S1	O	71	ERCL	O
12	Vss	-	32	A8	I	52	Vss	-	72	LOAD	I
13	AD7	I/O	33	VDD	-	53	S2	O	73	VDD	-
14	AD6	I/O	34	ARS	I (HIGH)	54	S3	O	74	TMB	I
15	AD5	I/O	35	INT1	O (NC)	55	S4	O	75	SINT1	I (LOW)
16	AD4	I/O	36	INT2	O	56	S5	O	76	SINT2	I (LOW)
17	AD3	I/O	37	DSCAN	O	57	S6	O	77	TEST1	I (LOW)
18	AD2	I/O	38	ERAM	O	58	S7	O	78	CLK	I
19	AD1	I/O	39	ERAM	O (NC)	59	R0	I	79	TEST2	I (LOW)
20	AD0	I/O	40	RS	O	60	R1	I	80	SYNT1	O

Target Custom Gate Array – Memory card controller

GATE ARRAY

μ PD65005G-062

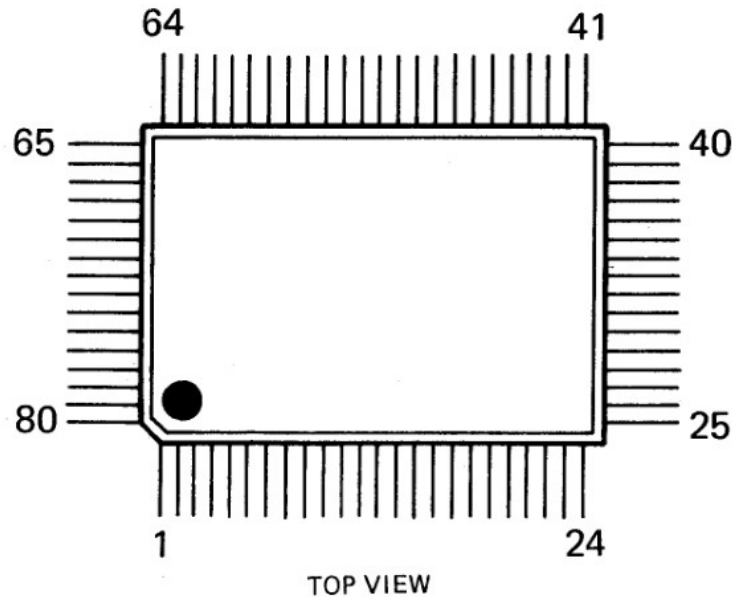


PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O
1	NC	-	17	NC	-	33	NC	-	49	NC	-
2	NC	-	18	NC	-	34	NC	-	50	CD0	I/O
3	AD7	I/O	19	A13	I	35	CA5	O	51	CD1	I/O
4	AD6	I/O	20	A12	I	36	CA6	O	52	CD2	I/O
5	AD5	I/O	21	A11	I	37	CA7	O	53	CD3	I/O
6	AD4	I/O	22	A10	I	38	CA8	O	54	CD4	I/O
7	AD3	I/O	23	A9	I	39	CA9	O	55	CD5	I/O
8	AD2	I/O	24	A8	I	40	CA10	O	56	CD6	I/O
9	AD1	I/O	25	SEL	I (LOW)	41	CA11	O	57	CD7	I/O
10	AD0	I/O	26	Vss	-	42	CA12	O	58	Vss	-
11	Vss	-	27	VDD	-	43	CA13	O	59	VDD	-
12	VDD	-	28	CA0	O	44	CA14	O	60	BATT	I (LOW)
13	ALE	I	29	CA1	O	45	MR	O	61	SENS	I (NC)
14	$\overline{\text{WR}}$	I	30	CA2	O	46	CWR	O	62	RCS	I
15	$\overline{\text{RD}}$	I	31	CA3	O	47	CCS	O	63	$\overline{\text{CS}}$	I
16	A14	I	32	CA4	O	48	CRD	O	64	NC	-

Target Custom Gate Array

– Keyboard controller

**GATE ARRAY
MB63H149**



PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O
1	T7	O	21	BR9	I	41	AD7	I/O	61	RA1	O
2	BR0	I	22	MK9	I	42	CA8	I	62	RA10	O
3	MK0	I	23	BR10	I	43	CA9	I	63	RA2	O
4	BR1	I	24	MK10	I	44	CA10	I (LOW)	64	$\overline{ROE}$	I/O
5	MK1	I	25	RES	I	45	CS	I	65	RA3	O
6	BR2	I	26	$\overline{EXCK}$	I/O	46	XT1	I	66	$\overline{RWE}$	O
7	MK2	I	27	E	I (HIGH)	47	XT2	O (NC)	67	RA4	O
8	BR3	I	28	INT	O	48	ASEL	O (NC)	68	RA9	O
9	MK3	I	29	AS	I	49	MOD1	I (HIGH)	69	RA5	O
10	BR4	I	30	$\overline{CRES}$	O (NC)	50	MOD2	I (LOW)	70	RA8	O
11	MK4	I	31	$\overline{CRNW}$	I	51	RD3	I/O	71	RA6	O
12	Vss	-	32	SRCK	O (NC)	52	Vss	-	72	RA7	O
13	BR5	I	33	VDD	-	53	RD4	I/O	73	VDD	-
14	MK5	I	34	AD0	I/O	54	RD2	I/O	74	T0	O
15	BR6	I	35	AD1	I/O	55	RD5	I/O	75	T1	O
16	MK6	I	36	AD2	I/O	56	RD1	I/O	76	T2	O
17	BR7	I	37	AD3	I/O	57	RD6	I/O	77	T3	O
18	MK7	I	38	AD4	I/O	58	RD0	I/O	78	T4	O
19	BR8	I	39	AD5	I/O	59	RD7	I/O	79	T5	O
20	MK8	I	40	AD6	I/O	60	RA0	O	80	T6	O